

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)

11
priors compared

32,276
held-out plate appearances

3.05%
held-out HR rate

0.6001
best AUC · KNN comps

served prior

Held-out scores

Fitted on plate appearances before 2026-08-28, scored on the 32,276 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5845	1.53×	0.02950	0.13547	0.00447
KNN comps Prior is the pooled rate of the nearest comparables.	0.6001	1.50×	0.02946	0.13452	0.00426
SVR Support-vector regression on the same feature matrix.	0.5974	1.48×	0.02949	0.13504	0.00614
Random forest Forest regression; a learned similarity metric.	0.5966	1.44×	0.02949	0.13488	0.00607
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6001	1.50×	0.02946	0.13452	0.00426
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6001	1.50×	0.02946	0.13452	0.00426
Linear regression Ordinary least squares on the same features.	0.5963	1.48×	0.02948	0.13482	0.00416
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5973	1.53×	0.02950	0.13495	0.00630
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5964	1.51×	0.02948	0.13482	0.00494
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5966	1.46×	0.02950	0.13494	0.00619

BLEND WEIGHTS (FOREST VARIANT)

KNN 1.00 SVR 0.00 RF 0.00

SHRINKAGE K PER PRIOR

league 248 knn 248 svr 248 rf 248 core 248 ensemble 248
linear 248 logistic 248 nn 248 xgb 248

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,511 PA, actual HR rate 2.36%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5561	1.45×	0.02311	0.11231	0.01006
KNN comps Prior is the pooled rate of the nearest comparables.	0.5834	1.33×	0.02314	0.11213	0.01242
SVR Support-vector regression on the same feature matrix.	0.5842	1.33×	0.02328	0.11503	0.01762
Random forest Forest regression; a learned similarity metric.	0.5926	1.21×	0.02329	0.11334	0.01709
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5834	1.33×	0.02314	0.11213	0.01242
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5834	1.33×	0.02314	0.11213	0.01242
Linear regression Ordinary least squares on the same features.	0.5863	1.08×	0.02322	0.11356	0.01512
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5910	1.21×	0.02323	0.11308	0.01480
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5854	1.21×	0.02318	0.11274	0.01443
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6006	1.21×	0.02330	0.11334	0.01711

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Ben Rice 21.9%, Munetaka Murakami 19.2%, Luis García Jr. 18.6%, Matt Olson 17.0%, Shohei Ohtani 16.9%, Fernando Tatis Jr. 16.6%
Flat league	4/6	Ben Rice 20.4%, Luis García Jr. 18.0%, Munetaka Murakami 17.6%, Fernando Tatis Jr. 16.6%, Ronald Acuña Jr. 15.6%, Jackson Chourio 15.4%
KNN comps	6/6	Ben Rice 21.7%, Munetaka Murakami 19.2%, Luis García Jr. 18.9%, Fernando Tatis Jr. 17.0%, Shohei Ohtani 16.9%, Matt Olson 16.5%
SVR	6/6	Ben Rice 22.4%, Munetaka Murakami 19.8%, Luis García Jr. 19.7%, Matt Olson 17.1%, Shohei Ohtani 17.1%, Fernando Tatis Jr. 16.9%
Random forest	6/6	Ben Rice 22.1%, Munetaka Murakami 19.4%, Luis García Jr. 19.1%, Shohei Ohtani 17.2%, Matt Olson 17.0%, Fernando Tatis Jr. 16.5%
Core (KNN+SVR)	6/6	Ben Rice 21.7%, Munetaka Murakami 19.2%, Luis García Jr. 18.9%, Fernando Tatis Jr. 17.0%, Shohei Ohtani 16.9%, Matt Olson 16.5%
Forest blend	6/6	Ben Rice 21.7%, Munetaka Murakami 19.2%, Luis García Jr. 18.9%, Fernando Tatis Jr. 17.0%, Shohei Ohtani 16.9%, Matt Olson 16.5%
Linear regression	6/6	Ben Rice 21.8%, Munetaka Murakami 19.0%, Luis García Jr. 19.0%, Fernando Tatis Jr. 17.2%, Shohei Ohtani 16.8%, Matt Olson 16.7%
Logistic regression	6/6	Ben Rice 22.1%, Luis García Jr. 19.4%, Munetaka Murakami 19.3%, Fernando Tatis Jr. 17.3%, Matt Olson 17.3%, Shohei Ohtani 17.3%
Neural net (torch)	6/6	Ben Rice 22.0%, Luis García Jr. 19.4%, Munetaka Murakami 19.3%, Fernando Tatis Jr. 17.3%, Shohei Ohtani 17.1%, Matt Olson 16.9%
XGBoost	6/6	Ben Rice 22.1%, Munetaka Murakami 19.7%, Luis García Jr. 19.3%, Shohei Ohtani 17.4%, Matt Olson 16.9%, Fernando Tatis Jr. 16.6%

Hitter by hitter

The top 40 bats by the served probability.

HITTER		SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (T)
19William Contreras	ATL	13.4%	13.3%	13.5%	13.2%	13.2%	13.5%	13.5%	13.3%	13.3%	
20Junior Camineró	TB	13.3%	12.5%	13.1%	14.5%	13.5%	13.1%	13.1%	13.3%	13.7%	
21Kyle Teel	CWS	13.2%	13.1%	13.7%	13.0%	13.4%	13.7%	13.7%	13.5%	12.6%	
22Jonathan Aranda	AB	13.2%	12.4%	13.5%	13.1%	13.2%	13.5%	13.5%	13.2%	13.0%	
23Drake Baldwin	ATL	13.0%	12.7%	13.2%	13.5%	13.8%	13.2%	13.2%	13.5%	13.3%	
24Jo Adell	CLE	12.9%	12.3%	12.7%	13.3%	12.9%	12.7%	12.7%	12.9%	13.2%	
25George Lombardi Jr.	CWS	12.9%	13.2%	12.8%	12.5%	13.0%	12.8%	12.8%	13.0%	12.4%	
26Michael Harris III	ATL	12.7%	12.8%	12.7%	13.2%	12.6%	12.7%	12.7%	12.9%	13.2%	
27Enrique Hernández	DET	12.6%	12.3%	12.4%	12.1%	12.3%	12.4%	12.4%	12.5%	12.0%	
28Randal Grichuk	CWS	12.1%	10.9%	12.0%	12.8%	11.9%	12.0%	12.0%	11.9%	12.8%	
29Jake Bauers	MIL	12.0%	11.2%	12.3%	12.2%	12.1%	12.3%	12.3%	12.1%	12.0%	
30Ty France	SD	12.0%	11.7%	12.3%	12.2%	12.2%	12.3%	12.3%	12.3%	12.2%	
31Austin Riley	ATL	12.0%	11.0%	11.8%	11.7%	11.5%	11.8%	11.8%	11.7%	11.8%	
32Colson Montgomery	CWS	11.9%	10.8%	12.0%	11.6%	12.0%	12.0%	12.0%	12.3%	12.8%	
33Christian Yelich	MIL	11.8%	12.0%	12.1%	11.5%	11.9%	12.1%	12.1%	12.3%	11.8%	
34Victor Mesa Jr.	TB	11.5%	11.4%	11.7%	12.0%	11.8%	11.7%	11.7%	11.8%	11.8%	
35Ozzie Albies	ATL	10.9%	11.3%	10.9%	11.5%	11.5%	10.9%	10.9%	11.2%	11.1%	
36Andy Pages	LAD	10.6%	10.1%	10.3%	10.0%	10.3%	10.3%	10.3%	10.1%	10.0%	
37Austin Hays	SD	10.6%	10.3%	10.0%	9.7%	9.7%	10.0%	10.0%	10.1%	9.4%	

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (T
38Tristan PetersCWS	10.5%	11.1%	10.6%	11.0%	11.0%	10.6%	10.6%	10.8%	10.9%	
39Mike YastrzemskiL	10.4%	10.4%	10.4%	10.3%	10.3%	10.4%	10.4%	10.1%	10.1%	
40Kyle TuckerLAD	10.4%	10.5%	10.4%	10.3%	10.4%	10.4%	10.4%	10.3%	10.1%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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Built for research and entertainment. Probabilities are estimates, not guarantees; never stake what you cannot afford to lose. Data: MLB Stats API play-by-play, lineups and live scores; ESPN injuries and bios.